

11.6 ELECTRIC POTENTIAL

The work done in displacing unit charge between a point and infinity is termed as electric potential at that point.

The electrical potential at a point is the electric potential difference between that point and infinity. Since electric potential at infinity is assumed to be zero, the difference of potential in this case gives the absolute potential at that point.

The unit of electric potential is $\frac{\text{Joule}}{\text{Coulomb}}$ which is also known as **volt**.

11.6.1 Electric Potential Energy

The work done by external agency in making a charged particle move against the force exerted by the electric field is stored as electric potential energy of the charge or charge particle. Now consider the process of assembling number of charges $q_1, q_2, q_3, \dots$. The process can be carried out as follows. Initially consider all the charges all the charges are at infinity. First bring the charge q_1 from infinity to a location $\vec{r}_1$. No work will be done to bring this charge as it is brought to a point where there is no electric field.

Now bring the charge q_2 from infinity to a location $\vec{r}_2$. Due to charge q_1 the electric potential at $\vec{r}_2$ is $\frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\vec{r}_2 - \vec{r}_1|}$ and hence work done in putting the charge q_2 at $\vec{r}_2$ is

$$U_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\vec{r}_2 - \vec{r}_1|}$$

Now if we bring the charge q_3 from infinity to $\vec{r}_3$ then potential at $\vec{r}_3$ due to charges q_1 and q_2 are

$$\frac{1}{4\pi\epsilon_0} \frac{q_1}{|\vec{r}_3 - \vec{r}_1|} \text{ and } \frac{1}{4\pi\epsilon_0} \frac{q_2}{|\vec{r}_3 - \vec{r}_2|}$$

and the total work done in putting the charge q_3 at $\vec{r}_3$ is sum of these two terms,

i.e.,
$$\frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{|\vec{r}_3 - \vec{r}_1|} \text{ and } \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{|\vec{r}_3 - \vec{r}_2|}$$

$\therefore$ The total work done in assembling the charges $q_1, q_2, q_3, \dots, q_n$ is

$$\begin{aligned} U_{123\dots n} &= \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\vec{r}_2 - \vec{r}_1|} + \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{|\vec{r}_3 - \vec{r}_1|} + \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{|\vec{r}_3 - \vec{r}_2|} + \dots \\ &= \sum_{i=1}^n \sum_{j=1}^n \frac{1}{4\pi\epsilon_0} \frac{q_j q_i}{|\vec{r}_j - \vec{r}_i|} \end{aligned}$$

where $i \neq j$. Here $U_{123\dots n}$ is called the potential energy of the system of charges.

Since the electric potential energy is measured in terms of the work done in displacing a charged particle in the electric field. Therefore, we can measure only the difference in the potential energies when the particle is located at the two positions. To carry out absolute measurement of potential energy, we arbitrarily assume that the potential energy of a charge is zero when it is located at infinite distance from the source of the electric field.

11.8 PHYSICAL SIGNIFICANCE OF DIVERGENCE

The divergence of vector field $\vec{E}$ ($\text{div } \vec{E}$) is defined as the limiting value of the ratio of the closed surface integral to the volume enclosed by the surface over which integration is carried out, when the volume tends to zero, i.e.,

$$\text{div } \vec{E} = \lim_{v \rightarrow 0} \frac{1}{v} \oiint \vec{E} \cdot d\vec{s}$$

where v is the volume enclosed by the source S over which integration is carried out. In vector form $\text{div } \vec{E}$ is represented by $\vec{\nabla} \cdot \vec{E}$.

The divergence of a vector field $\vec{E}$ can be expressed as

$$\text{div } \vec{E} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (\hat{i} E_x + \hat{j} E_y + \hat{k} E_z) = \vec{\nabla} \cdot \vec{E}$$

where symbol $\vec{\nabla}$ (del) is a vector differential operator and is given by $\vec{\nabla} = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$.

The divergence of electric field at any point gives the charge density at that point, which is scalar quantity.

If a vector function $\vec{E}$ spreads out, i.e., diverges from a point, then it has a positive divergence at that point and point acts as a source of the field $\vec{E}$. Indeed $\vec{\nabla} \cdot \vec{E}$ can be taken to be a measure of the spreading out of the field. On the other hand, if the field converges to a point then $\vec{\nabla} \cdot \vec{E}$ will be negative at that point because the point acts as a sink for the field $\vec{E}$. Finally, if the vector field $\vec{E}$ neither converges nor diverges then $\vec{\nabla} \cdot \vec{E} = 0$, i.e., the flux entering any element of space is same as leaving it and such a vector field is known as **solenoidal vector field**.

11.9 PHYSICAL SIGNIFICANCE OF CURL

The curl of a vector field signifies the whirling nature of the vector field which sometime also known as rotation. It is directed along the perpendicular to the plane of maximum rotation which can be proved by considering the streamline flow of a liquid in a pipe. There exists a velocity gradient in the liquid and its velocity is maximum at the top and minimum at the bottom. Put a toothed wheel, with its plane along the direction of flow. It will begin to rotate in the direction of flow. Thus, the rotation of the wheel in this position will be maximum and it will be minimum if the toothed wheel is put with its plane perpendicular to the direction of flow. The rotation in any intermediate position will be in between the maximum and minimum value.

The rotation with maximum value is termed as curl and is a vector quantity. If we denote the flow field in the above example by $\vec{B}$, then rotation may be represented as $\text{curl } \vec{B}$. It is represented in accordance with the right handed screw rule, e.g. in the above case, the rotation being in clockwise direction, represents the curl into the plane of the paper (Fig. 11.4).

The curl of a vector field $\vec{B}$ can be expressed as $\text{curl } \vec{B} = \vec{\nabla} \times \vec{B}$.

Non-zero curl of a vector field implies existence of circulation or velocity or rotation. It suggests a whirling effect or the formation of a vortex. We have seen the vortex in the river or in the tub, when water is allowed to flow out of it through a hole in the bottom Fig. 11.5. $\vec{\nabla} \times \vec{E} = 0$ signifies that there is no tendency of rotation in an electrostatic field.

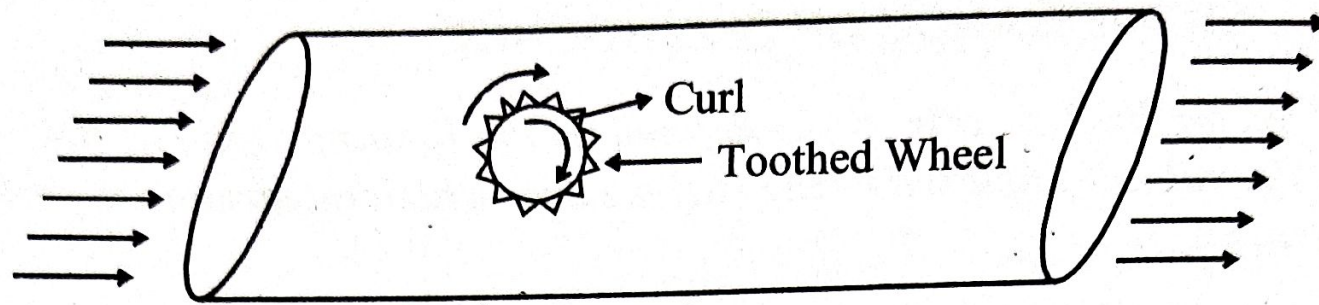


Fig. 11.4 Clockwise rotation of Toothed Wheel in the Direction of Flow

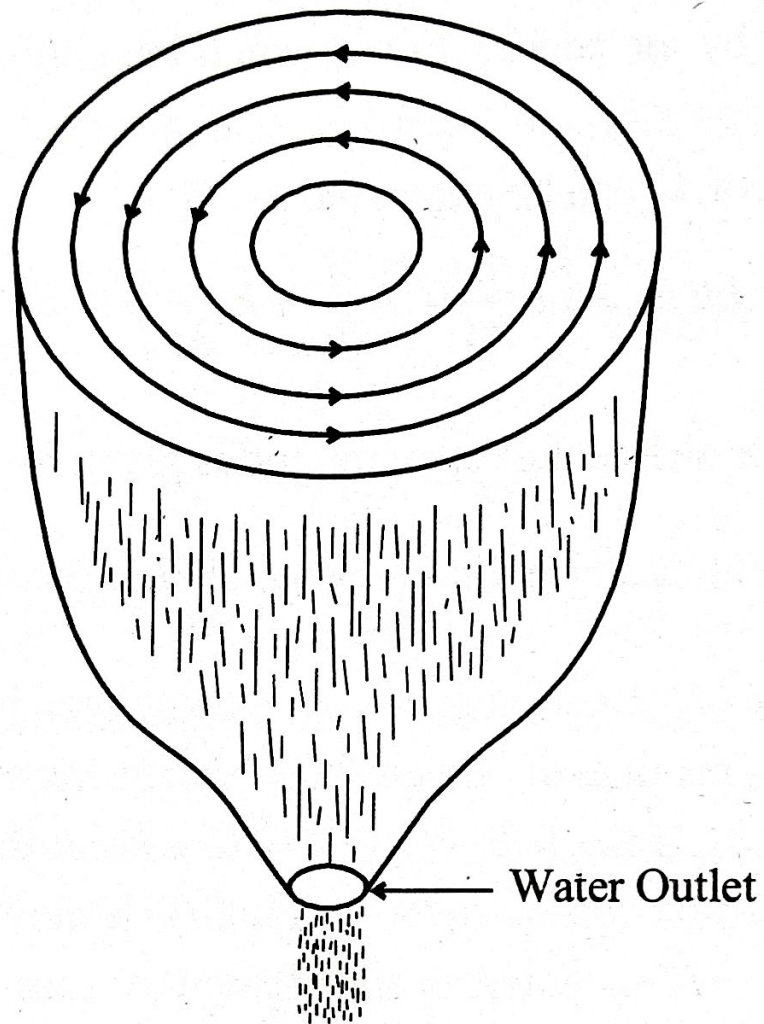


Fig. 11.5 Demonstration of Vortex

11.11 RELATIONSHIP BETWEEN ELECTRIC FIELD AND ELECTRIC POTENTIAL

We have seen that the electric potential is given by

$$U(\vec{r}) = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r}$$

Now take gradient of above expression, we have

$$\vec{\nabla}U = \vec{\nabla}\left(\frac{1}{4\pi\epsilon_0} \frac{q}{r}\right) = \frac{1}{4\pi\epsilon_0} q \vec{\nabla}\left(\frac{1}{r}\right)$$

Since, $\vec{r} = \hat{i}x + \hat{j}y + \hat{k}z$, we have

$$\begin{aligned}\vec{\nabla}\left(\frac{1}{r}\right) &= \vec{\nabla}(x^2 + y^2 + z^2)^{-\frac{1}{2}} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}\right)(x^2 + y^2 + z^2)^{-\frac{1}{2}} \\ &= -\frac{1}{2}(x^2 + y^2 + z^2)^{-3/2} (\hat{i} 2x + \hat{j} 2y + \hat{k} 2z) \\ &= -r^{-3} \vec{r} = -\frac{1}{r^2} \frac{\vec{r}}{r} = -\frac{\hat{r}}{r^2} \quad (\because \vec{r} = |\vec{r}| \hat{r})\end{aligned}$$

$$\therefore \vec{\nabla}U = \frac{1}{4\pi\epsilon_0} q \left(\frac{-\hat{r}}{r^2}\right) = -\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

or

$$\vec{\nabla}U = -\vec{E}$$

$$\left(\because \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}\right)$$

or

$$\vec{E} = -\vec{\nabla}U$$

...(3)

Thus, electric field at a point is defined as the gradient of the potential at that point.

11.11.1 Curl of Electric Field $\vec{E}$ is Zero

Since

$$\vec{E} = -\nabla U$$

Therefore,

$$\text{Curl } \vec{E} = \vec{\nabla} \times \vec{E} = \vec{\nabla} \times [-\vec{\nabla} U]$$

i.e.,

$$\begin{aligned} \vec{\nabla} \times \vec{E} &= - \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial U}{\partial x} & \frac{\partial U}{\partial y} & \frac{\partial U}{\partial z} \end{vmatrix} \\ &= - \left[\hat{i} \left(\frac{\partial^2 U}{\partial z \partial y} - \frac{\partial^2 U}{\partial y \partial z} \right) + \hat{j} \left(\frac{\partial^2 U}{\partial x \partial z} - \frac{\partial^2 U}{\partial z \partial x} \right) + \hat{k} \left(\frac{\partial^2 U}{\partial x \partial y} - \frac{\partial^2 U}{\partial y \partial x} \right) \right] = 0. \end{aligned}$$

That is, curl of electric field is zero.

11.12 DIV. GRAD U ; LAPLACIAN OPERATOR

$$\begin{aligned} \text{div. grad } U &= \vec{\nabla} \cdot \vec{\nabla} U = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \left(\hat{i} \frac{\partial U}{\partial x} + \hat{j} \frac{\partial U}{\partial y} + \hat{k} \frac{\partial U}{\partial z} \right) \\ &= \frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) U \end{aligned}$$

$$\therefore \vec{\nabla} \cdot \vec{\nabla} U = \vec{\nabla}^2 U$$

where $\vec{\nabla}^2 \left(= \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right)$ is known as Laplacian operator.

11.13 DIFFERENTIAL VECTOR IDENTITIES

Followings are the possible combinations of differential operators and products which can be applied to various products of two vectors (e.g., $\vec{E}$ and $\vec{E}'$) and scalars (e.g., U).

$$(i) \quad \vec{\nabla} \cdot \vec{\nabla} U = \vec{\nabla}^2 U$$

$$(ii) \quad \vec{\nabla} \cdot \vec{\nabla} \times \vec{E} = 0$$

$$(iii) \quad \vec{\nabla} \times \vec{\nabla} U = 0$$

$$(iv) \quad \vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{E}) - \vec{\nabla}^2 \vec{E}$$

$$(v) \quad \vec{\nabla} (U \vec{E}) = (\vec{\nabla} U) \vec{E} + U \vec{\nabla} \cdot \vec{E}$$

$$(vi) \quad \vec{\nabla} (\vec{E} \times \vec{E}') = (\vec{E} \cdot \vec{\nabla}) \vec{E}' + \vec{E} \times (\vec{\nabla} \times \vec{E}') + (\vec{E}' \cdot \vec{\nabla}) \vec{E} + \vec{E}' \times (\vec{\nabla} \times \vec{E})$$

$$(vii) \quad \vec{\nabla} \cdot (U \vec{E}) = (\vec{\nabla} U) \cdot \vec{E} + U \vec{\nabla} \cdot \vec{E}$$

$$(viii) \quad \vec{\nabla} \cdot (\vec{E} \times \vec{E}') = (\vec{\nabla} \times \vec{E}) \cdot \vec{E}' - (\vec{\nabla} \times \vec{E}') \cdot \vec{E}$$

$$(ix) \quad \vec{\nabla} \times (U \vec{E}) = (\vec{\nabla} U) \times \vec{E} + U \vec{\nabla} \times \vec{E}$$

$$(x) \quad \vec{\nabla} \times (\vec{E} \times \vec{E}') = (\vec{\nabla} \cdot \vec{E}') \vec{E} - (\vec{\nabla} \cdot \vec{E}) \vec{E}' + (\vec{E}' \cdot \vec{\nabla}) \vec{E} - (\vec{E} \cdot \vec{\nabla}) \vec{E}'$$

11.14 TYPES OF VECTOR FIELDS

1. Lamellar Vector Field (LVF): A vector field is said to be lamellar (Laminar) if it can be expressed as the gradient of the a scalar field. Electric field is a lamellar field as it is expressed by gradient of potential U

i.e.,
$$\vec{E} = -\vec{\nabla}U$$

The name lamellar suggests that the field can be divided into layers over which the value of the scalar function whose gradient gives the vector field, remains constant.

The line integral of a lamellar vector is independent of the path followed and only depends on the initial and final points of the path, e.g., the line integral of a lamellar field between two points A and B is given by

$$\int_A^B \vec{E} \cdot d\vec{l} = \int_A^B -\vec{\nabla}U \cdot d\vec{l} = - \int_A^B dU = U_A - U_B$$

where U is the electric potential and U_A and U_B are values of electric potential at A and B respectively.

From last equation, we may write

$$\oint \vec{E} \cdot d\vec{l} = \int_A^B \vec{E} \cdot d\vec{l} = U_A - U_B = 0$$

i.e., the closed line integral of a lamellar field is zero.

2. Solenoidal Vector Field (SVF): A vector field $\vec{B}$ is said to be solenoidal if its divergence is zero.

i.e.,
$$\vec{\nabla} \cdot \vec{B} = 0 \quad \dots(4)$$

Solenoidal fields neither have source nor sink of flux and the flux lines entering a closed surface must also leave it. Incompressible fluids, steady magnetic fields, current density under steady state are some of the examples of solenoidal vector fields.

3. Irrotational Vector Field (IVF): A vector field $\vec{E}$ whose curl is zero is called irrotational vector field.

i.e.,
$$\vec{\nabla} \times \vec{E} = 0 \quad \dots(5)$$

This equation is satisfied only by electric field $\vec{E}$.

$$(\because \vec{E} = -\vec{\nabla}U \text{ then } \vec{\nabla} \times \vec{E} = \vec{\nabla} \times [-\vec{\nabla}U] = -[\vec{\nabla} \times \vec{\nabla}U])$$

$$\text{But } (\vec{\nabla} \times \vec{\nabla} = 0 \therefore \vec{\nabla} \times \vec{\nabla}U = 0 \text{ or } \vec{\nabla} \times \vec{E} = 0)$$

4. Rotational Vector Field (RVF): A vector field whose curl is non-zero is called rotational vector field. Magnetic flux density $\vec{B}$ satisfies this condition and is said to be rotational.

i.e.,
$$\vec{\nabla} \times \vec{B} \neq 0 \quad \dots(6)$$